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A Project Report on

**“RISK ADAPTIVE VOICE-BASED
FINANCE ASSISTANT”**

Submitted in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

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DECLARATION

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Signature of Student(s)

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ABSTRACT

Voice-based financial transactions are a new paradigm of digital banking, but they introduce new security issues at the crossroads of speech recognition, biometric authentication, and fraud detection. This paper describes a Risk-Adaptive Voice Based Financial Assistant, which uses a multi-modal machine learning pipeline consisting of four key modules: (1) Speech-to Text transcription with OpenAI Whisper, (2) Speaker Verification with ECAPA-TDNN embeddings with cosine similarity and anti spoofing liveness detection, (3) Intent Classification with a Bidirectional LSTM network. The system also presents a new three-layered, risk-adaptive authentication scheme that dynamically chooses the authentication scheme, which can be PIN-only, Step Up (PIN + voice reconfirmation), or Hard Block, depending on a composite risk score based on similarity with speaker verification, fraud anomaly scores, and context of the transaction. Experimental findings indicate that the intent classifier has a weighted F1-score of 0.995, the speaker verification module has an Equal Error Rate (EER) of less than 2% with the enrolled speakers and the end-to end pipeline has a processing time of less than 200ms with the CPU. The system is tested with a complete implementation of React frontend, FastAPI backend, and Razorpay payment gateway.

Keywords—voice biometrics, speaker verification, ECAPA TDNN, intent classification, LSTM, fraud detection, isolation forest, risk-adaptive authentication, anti-spoofing, Whisper STT

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CHAPTER 1

1.1 INTRODUCTION

Voice-enabled interfaces are becoming an important part of modern digital systems. Applications such as virtual assistants, voice search, and voice-enabled smart devices are widely used in daily life. In financial systems, voice commands can make digital payments easier, faster, and more accessible.

However, voice-based financial transactions introduce many security challenges. A voice payment system must not only understand the spoken command correctly but also verify that the person speaking is the authorized user. In addition, the system must identify suspicious or fraudulent transaction patterns.

Traditional systems use static security mechanisms such as passwords and PINs. These methods are vulnerable because an attacker can steal credentials or replay a recorded voice. Therefore, there is a need for a secure, adaptive, and intelligent voice-based payment system.

The proposed project addresses this issue by integrating multiple machine learning models into a single multi-modal pipeline. The system can transcribe speech, classify user intent, verify the speaker, detect spoofing attacks, and identify fraud risks before approving a transaction.

1.2 MOTIVATION

The growth of UPI and mobile banking in India has increased the demand for faster and more user-friendly payment methods. Elderly people, visually impaired users, and users unfamiliar with mobile applications may find voice-based transactions more convenient.

At the same time, financial institutions require strong security mechanisms to prevent unauthorized access and fraud. Therefore, the system must ensure both usability and security.

1.3 AIM AND OBJECTIVE OF THE WORK

Aim:

To develop a secure and intelligent voice-based financial assistant that performs speech recognition, speaker verification, fraud detection, and adaptive authentication for safe digital transactions.

Objectives:

- To build a secure voice-based payment assistant.
- To convert spoken financial commands into text.
- To identify the user's transaction intent.
- To verify the identity of the speaker.
- To detect replay attacks and spoofing attempts.
- To calculate fraud risk based on user behaviour.
- To provide adaptive authentication based on risk.

CHAPTER 2

2.1 LITERATURE SURVEY

2.1.1 Desplanques et al. proposed the ECAPA-TDNN architecture for speaker verification. The model stands for Emphasized Channel Attention, Propagation and Aggregation in Time Delay Neural Network. It improves the traditional TDNN architecture by introducing channel attention and multi-scale feature aggregation. The ECAPA-TDNN model generates 192-dimensional speaker embeddings that capture unique voice characteristics of a person. The model achieved state-of-the-art performance on the VoxCeleb dataset with very low Equal Error Rate. The main advantage of this model is that it can accurately verify speakers even with very short voice samples of about 3 to 5 seconds. This makes it highly suitable for financial applications where the user only speaks a short command such as “Send 500 rupees to Ravi”.

In the proposed project, ECAPA-TDNN is used for speaker verification because it offers better accuracy than older x-vector and TDNN models.

2.1.2 Radford et al. developed Whisper, a robust automatic speech recognition model trained on approximately 680,000 hours of multilingual and multitask supervised data collected from the web. Whisper is based on a transformer architecture and is capable of handling different accents, background noises, and languages.

Unlike traditional speech recognition systems, Whisper performs well even when there is environmental noise, unclear pronunciation, or mixed-language speech. This is important in Indian financial applications because users may speak in English mixed with regional languages, and there may be noise from traffic, television, or surrounding people.

The Whisper model converts spoken commands into text, which is later passed to the intent classification module. Because of its high accuracy and robustness, Whisper was selected in this project for speech-to-text conversion.

2.1.3 Liu et al. proposed the Isolation Forest algorithm for anomaly detection. The basic idea behind the model is that anomalies are easier to isolate than normal data points. The model randomly partitions data using decision trees and calculates an anomaly score for each observation.

In financial systems, fraudulent transactions usually behave differently from normal user behaviour. For example, a transaction with a very large amount, a new recipient, or an unusual transaction time may indicate fraud. Isolation Forest can identify such unusual patterns quickly and efficiently.

The advantage of Isolation Forest is that it works well even without labeled fraud data. It is also computationally efficient and suitable for real-time systems. Therefore, it is used in the proposed project to detect suspicious financial activities.

2.1.4 Chen and Guestrin proposed the use of ensemble learning methods for improving prediction accuracy in machine learning systems. Their work showed that combining multiple models often gives better results than using a single model alone.

In fraud detection, supervised models such as Random Forest are good at detecting known fraud patterns, while unsupervised models such as Isolation Forest are better at finding new or unusual behaviours.

By combining both models, the system becomes more accurate and robust.

The proposed project uses a fraud detection ensemble that combines Isolation Forest and Random Forest. The final fraud score is calculated by assigning 40% weight to Isolation Forest and 60% weight to Random Forest.

2.1.5 Kim proposed Bidirectional LSTM networks for text classification tasks. Unlike ordinary recurrent neural networks, BiLSTM can process input sequences in both forward and backward directions. This allows the model to capture more contextual information from a sentence.

For example, in a command such as “Send 500 rupees to Rahul”, the BiLSTM model can understand that “Send” is the action, “500 rupees” is the amount, and “Rahul” is the recipient. This improves the system’s ability to classify user intent correctly.

In the proposed system, BiLSTM is used to classify the user’s spoken command into one of the following categories:

- Send Money
- Check Balance

- Transaction History
- Pay Bill

The model achieved a weighted F1-score of 0.995, which shows that BiLSTM is highly effective for financial intent classification.

2.1.6 Shi et al. proposed techniques for live voice detection in speaker verification systems. Their work focused on distinguishing genuine live speech from replay attacks, where an attacker uses a recorded voice sample to fool the system.

Replay attacks often lose some high-frequency components because of microphone and speaker limitations during recording and playback. This property can be used to detect whether the input is genuine or spoofed.

In the proposed project, High Frequency Energy (HFE) ratio analysis is used as a simple anti-spoofing method. If the HFE ratio is below a certain threshold, the system identifies the audio as a replay attack and blocks the transaction.

2.1.7 Existing Voice Banking Systems

Several companies such as Google, Amazon, and Apple have developed voice assistants like Google Assistant, Alexa, and Siri. These systems can perform voice-based tasks such as checking information, controlling devices, and sending messages.

However, most commercial voice assistants are not designed for secure financial transactions. They may support voice commands for banking, but they usually depend only on passwords or device-level authentication. They do not combine speaker verification, fraud detection, and adaptive risk analysis.

Because of this limitation, there is a need for a more secure and intelligent voice-based financial system that can verify the user's identity, detect fraud, and adapt authentication based on risk level.

2.2 INFERENCES DRAWN FROM LITERATURE SURVEY

- ECAPA-TDNN provides high speaker verification accuracy for short voice samples.
- Whisper performs well in noisy and multilingual conditions.
- Isolation Forest can identify unusual financial transactions efficiently.
- Random Forest improves fraud detection when combined with anomaly detection methods.
- BiLSTM is highly effective for financial intent classification.
- Replay attacks can be identified using High Frequency Energy analysis.
- Existing commercial voice assistants do not provide complete security for financial transactions.
- A multi-modal machine learning pipeline provides better security, accuracy, and reliability than single-model systems.

Sr. No.	Paper Title	Author	Year	Method Used	Contribution
1	ECAPA-TDNN Based Speaker Verification	Desplanques et al.	2020	ECAPA-TDNN	Accurate speaker verification using channel attention and multi-scale feature aggregation
2	Whisper: Robust Speech Recognition via Large-Scale Weak Supervision	Radford et al.	2022	Transformer-based Speech-to-Text	Robust multilingual speech recognition in noisy environments
3	Isolation Forest	Liu et al.	2008	Anomaly Detection	Efficient detection of suspicious and unusual transaction behaviour
4	Ensemble Learning for Fraud Detection	Chen and Guestrin	2016	Ensemble Learning	Combines multiple models for better fraud prediction accuracy
5	Bidirectional LSTM for Intent Classification	Kim	2014	BiLSTM	Improved text classification by capturing forward and backward context
6	VoiceLive: Live Voice Detection for Speaker Verification	Shi et al.	2021	Anti-Spoofing	Detection of replay attacks using live voice verification
7	Commercial Voice Assistants for Banking Applications	Google, Amazon, Apple	2021	Voice Assistant Systems	Supports voice interaction but lacks advanced fraud detection and speaker verification

Table 1. Literature Survey Comparison

CHAPTER 3

SOFTWARE REQUIREMENT SPECIFICATION

3.1 INTRODUCTION

This Software Requirement Specification (SRS) document provides a detailed description of the proposed Risk-Adaptive Voice-Based Financial Assistant system. It explains the functional and non-functional requirements, system features, hardware and software requirements, and the analysis models used in the project.

The main objective of the system is to provide a secure and intelligent platform for voice-based financial transactions. The system integrates speech recognition, speaker verification, fraud detection, and adaptive authentication into a single framework.

3.1.1 Project Purpose

The purpose of the project is to design and develop a secure voice-based financial transaction assistant that can:

- Convert spoken commands into text.
- Identify the user's transaction intent.
- Verify the identity of the speaker.
- Detect fraudulent transactions.
- Prevent replay attacks.
- Approve or reject transactions based on risk.

The system is designed to reduce the dependency on traditional password-based authentication and improve both security and user convenience.

3.1.2 Project Scope

- The system is designed to support secure voice-based financial transactions.
- It can identify different financial intents such as sending money, checking balance, paying bills, and viewing transaction history.

- The system analyses the identity of the speaker and verifies whether the user is genuine or not.
- It can detect replay attacks and suspicious transactions.
- The project can be used by banking users, elderly users, visually impaired users, and users who prefer voice interaction over traditional applications.
- It can also be integrated with digital payment systems and banking applications for secure voice-based transactions.

3.2 FUNCTIONAL REQUIREMENT

Functional requirements describe the main operations and features of the system.

3.2.1 User Registration

- The system should allow new users to register.
- Users must provide their personal details and create a PIN.
- The system should record at least three voice samples for speaker enrollment.
- Speaker embeddings should be stored in the database.

3.2.2 User Login

- The system should allow users to log in securely.
- Users should be authenticated using voice and PIN.

3.2.3 Speech-to-Text Conversion

- The system should convert user voice commands into text using Whisper.
- The system should display the transcribed text for verification.

3.2.4 Intent Classification

- The system should classify the spoken command into different intents such as:
 - Send Money
 - Check Balance
 - Pay Bill
 - Transaction History

3.2.5 Speaker Verification

- The system should verify whether the speaker is the enrolled user.
- The system should compare the current speaker embedding with the stored embedding.

3.2.6 Fraud Detection

- The system should analyze transaction behaviour.
- The system should detect unusual transactions based on amount, recipient, time, and failed attempts.

3.2.7 Adaptive Authentication

- The system should choose one of the following authentication methods:
 - PIN Only
 - Step-Up Authentication
 - Transaction Block

3.2.8 Transaction Processing

- The system should execute valid transactions.
- The system should reject suspicious transactions.

3.3 NON-FUNCTIONAL REQUIREMENT

Non-functional requirements describe the quality and performance standards of the system.

3.3.1 Safety Requirements

- User data must be stored securely.
- Transactions must not be processed without proper authentication.
- Sensitive user information should not be exposed to unauthorized persons.
- The system should generate audit logs for all important activities.

3.3.2 Security Requirements

- User PINs must be stored using bcrypt hashing.
- Speaker embeddings should be encrypted.
- The system should protect against replay attacks.
- Unauthorized users should not be able to access transaction features.
- Fraud detection should be enabled for every transaction.

3.3.3 Software Quality Attributes

▪ Reliability

The system should provide consistent results without frequent failures.

▪ Accuracy

The system should correctly recognize speech, identify the speaker, and classify transaction intent.

▪ Performance

The system should process requests quickly and maintain low latency.

▪ Scalability

The system should support a large number of users and transactions.

▪ Usability

The user interface should be simple and easy to use.

▪ Maintainability

The system should be easy to update and modify in the future.

3.4 SYSTEM REQUIREMENT

3.4.1 Software Requirements

- Windows 10 and above
- Python 3.10 or above
- React JS
- FastAPI
- SQLite
- Visual Studio Code
- Whisper Model
- ECAPA-TDNN
- TensorFlow / PyTorch
- Scikit-learn
- Razorpay API Integration

3.4.2 Hardware Requirements

- 8 GB RAM
- Intel Core i5 / AMD Ryzen 5
- NVIDIA GeForce GTX 1650 or above
- CUDA Enabled GPU
- 256 GB SSD and 1 TB HDD
- Good Quality Microphone
- Stable Internet Connection

3.5 ANALYSIS MODELS

The project uses multiple machine learning and deep learning models for secure transaction processing.

3.5.1 Speech Recognition Model

Whisper is used to convert spoken audio into text.

3.5.2 Speaker Verification Model

ECAPA-TDNN is used to generate speaker embeddings and compare them with enrolled user voice samples.

3.5.3 Intent Classification Model

BiLSTM is used to classify user intent from the transcribed text.

3.5.4 Fraud Detection Model

Isolation Forest and Random Forest are used together to identify suspicious transactions.

3.5.5 Anti-Spoofing Model

High Frequency Energy ratio analysis is used to detect replay attacks.

CHAPTER 4

DESIGN AND MODELLING OF SYSTEM

4.1 Overall System Design

The proposed system is designed using a multi-layer architecture that combines speech recognition, intent classification, speaker verification, fraud detection, and adaptive authentication.

The system starts by taking a voice command from the user. The audio is then converted into text, analyzed for transaction intent, verified against the enrolled speaker profile, and checked for fraud risk before completing the transaction.

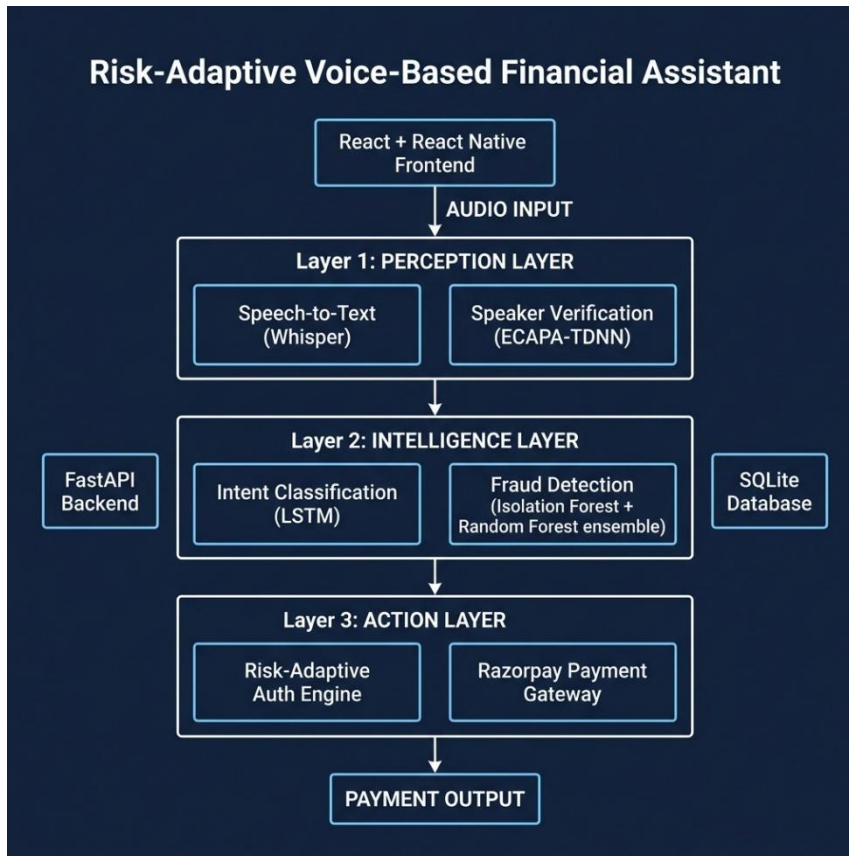


Fig1. System Architecture

This figure shows the complete architecture of the proposed system including frontend, backend, machine learning models, database, and payment gateway.

4.2 Voice Processing and Transaction Flow

The voice processing flow begins when the user records a voice command. The system sends the audio to the backend, where Whisper converts the speech into text.

The text is then passed to the intent classification module, which identifies the type of transaction. At the same time, the speaker verification module checks whether the voice belongs to the enrolled user.

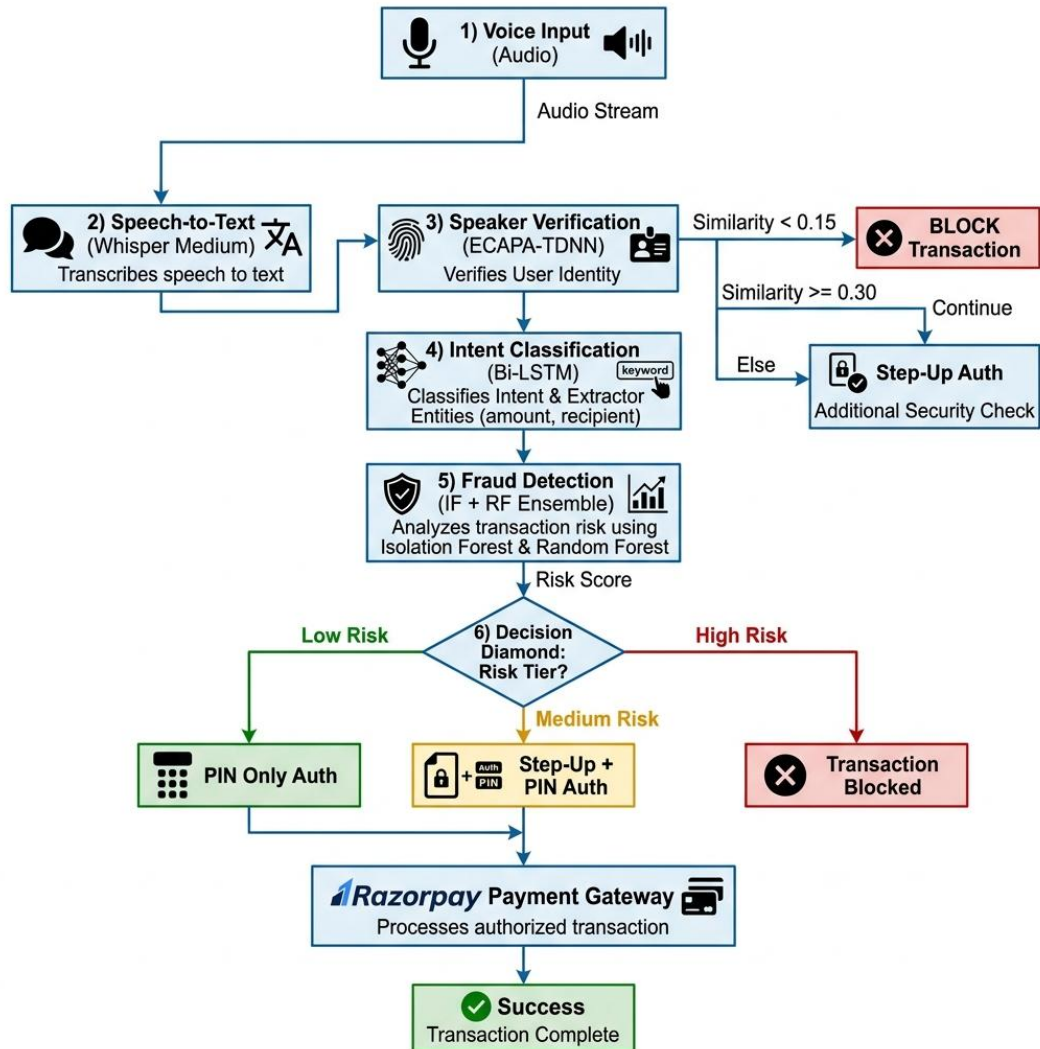


Fig2: Pipeline flow

This figure shows the flow from voice recording to speech recognition, intent classification, speaker verification, and transaction processing.

4.3 Speaker Verification Model

The speaker verification module uses ECAPA-TDNN embeddings to compare the current user voice with the stored enrolled voice sample.

The system calculates the cosine similarity score between the two embeddings. If the similarity is high, the user is treated as genuine. If the similarity is low, the transaction is blocked.

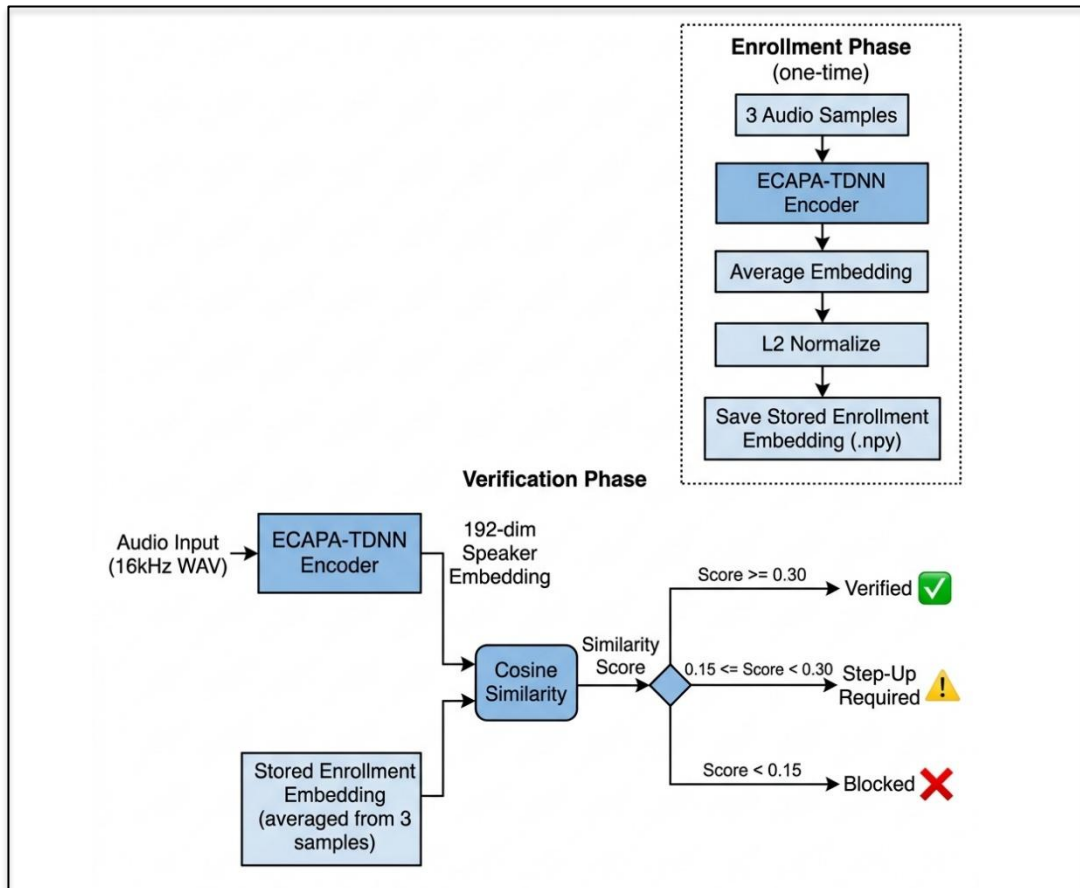


Fig3: Speaker Verification

This figure shows the similarity score distribution between genuine speakers and imposters.

4.4 Fraud Detection Model

The fraud detection model uses transaction features such as amount, transaction frequency, new recipient, failed authentication attempts, and transaction time.

These features are passed through Isolation Forest and Random Forest models. The outputs are combined to generate the final fraud risk score.

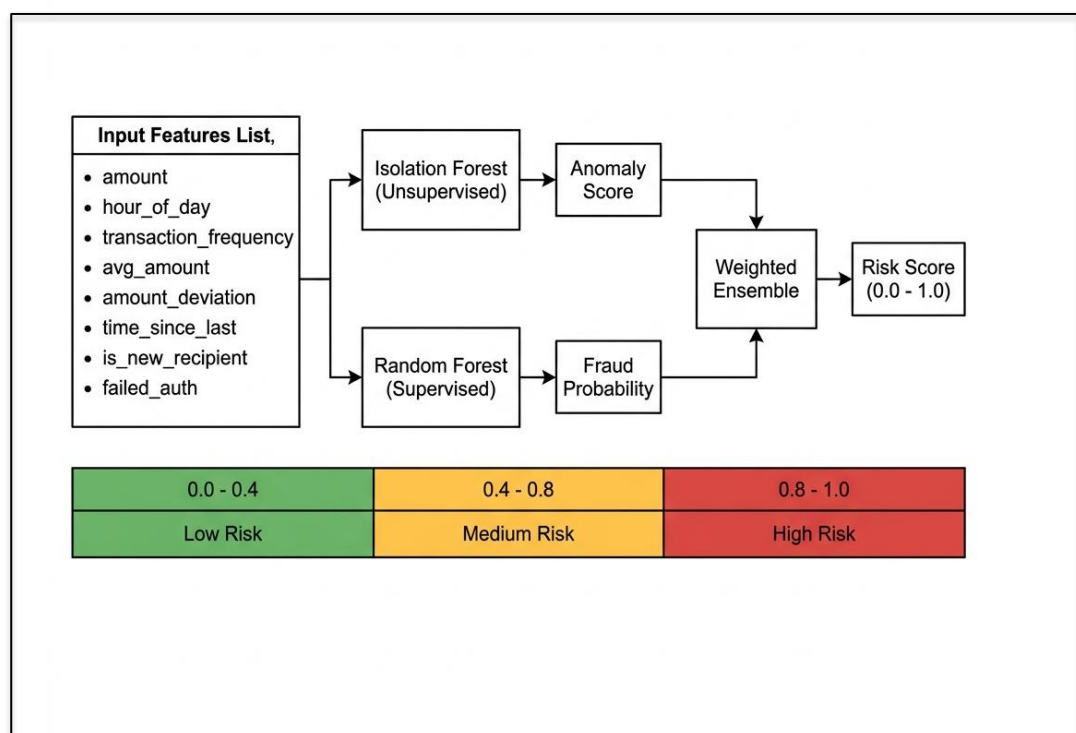


Fig4: Fraud ensemble

This figure shows how different transaction features are processed to generate the final fraud score.

4.5 Risk-Based Authentication Model

The final transaction decision depends on the calculated fraud score and speaker verification score.

- Low Risk → PIN Only
- Medium Risk → Step-Up Authentication
- High Risk → Transaction Block

Risk Tier Distribution (500 Test Transactions)

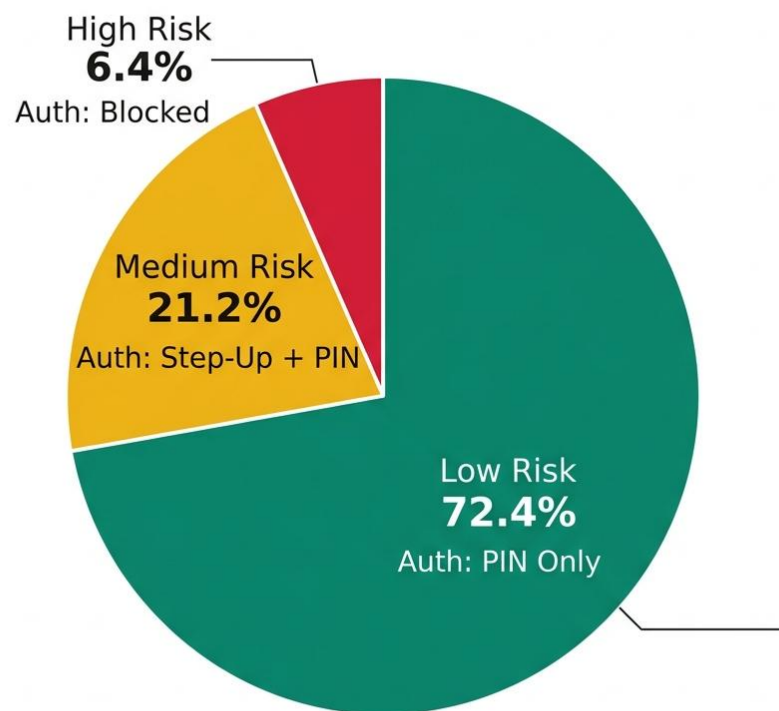


Fig5: Risk distribution

This figure shows how the system maps different risk levels to different authentication methods.

CHAPTER 5

ALGORITHM AND METHODS

5.1 Data Acquisition

The system collects voice samples from users during registration and transaction stages. During registration, the user is required to record multiple voice samples so that the system can create a speaker profile for future verification.

The audio is recorded in WAV format with a sampling rate of 16 kHz and mono channel configuration. The collected voice data is stored temporarily for feature extraction and processing.

The system also collects transaction-related features such as:

- Transaction amount
- Recipient name
- Time of transaction
- Transaction frequency
- Failed authentication attempts
- New or existing recipient

5.2 Data Pre-processing

Before processing, the recorded audio is cleaned and normalized. Noise reduction techniques are applied to improve speech quality.

The system extracts useful information from the audio signal such as:

- Spectrograms
- Mel Frequency Features
- Speaker Embeddings
- High Frequency Energy Ratio

The transcribed text is also pre-processed before being passed into the intent classification model. This includes:

- Converting text to lowercase

- Removing unwanted symbols
- Tokenizing words
- Padding sequences

5.3 Data Augmentation

To improve the model performance, different data augmentation methods are used during training.

For audio data, the following augmentation techniques are applied:

- Background noise addition
- Pitch shifting
- Time stretching
- Volume adjustment
- Random silence insertion

For text data, the following augmentation methods are used:

- Word dropout
- Filler word insertion
- Whisper-like transcription errors

These augmentation methods help the model perform well in real-world conditions where noise and speech variations are common.

5.4 Proposed Network

The proposed system combines multiple machine learning models in sequence.

1. Whisper Model for Speech-to-Text
2. BiLSTM for Intent Classification
3. ECAPA-TDNN for Speaker Verification
4. HFE Ratio for Anti-Spoofing
5. Isolation Forest and Random Forest for Fraud Detection

The speaker verification similarity is calculated using:

$$sim(e_{in}, e_{enr}) = \frac{e_{in} \cdot e_{enr}}{|| e_{in} || \cdot || e_{enr} ||}$$

The fraud score is calculated using:

$$R_{fraud} = 0.4R_{IF} + 0.6R_{RF}$$

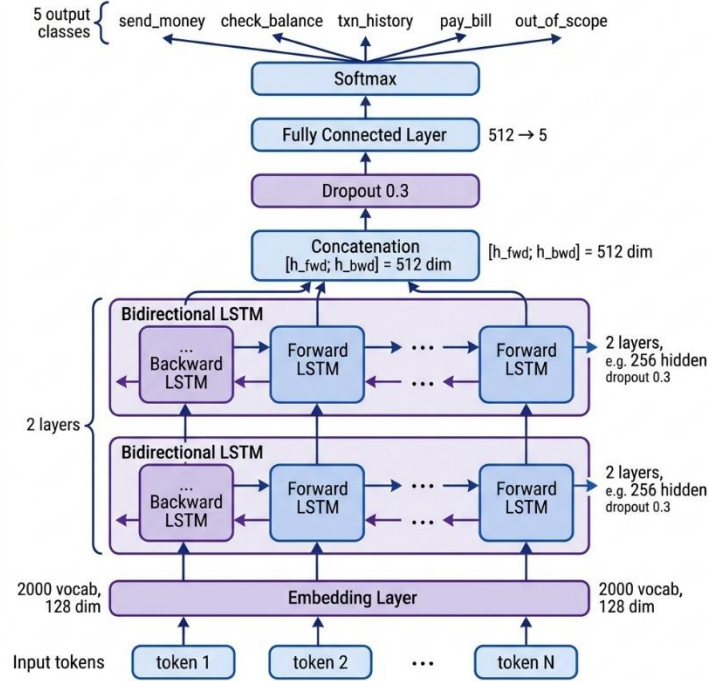


Fig6: BiLSTM Architecture

This figure shows the architecture of the BiLSTM model used for intent classification.

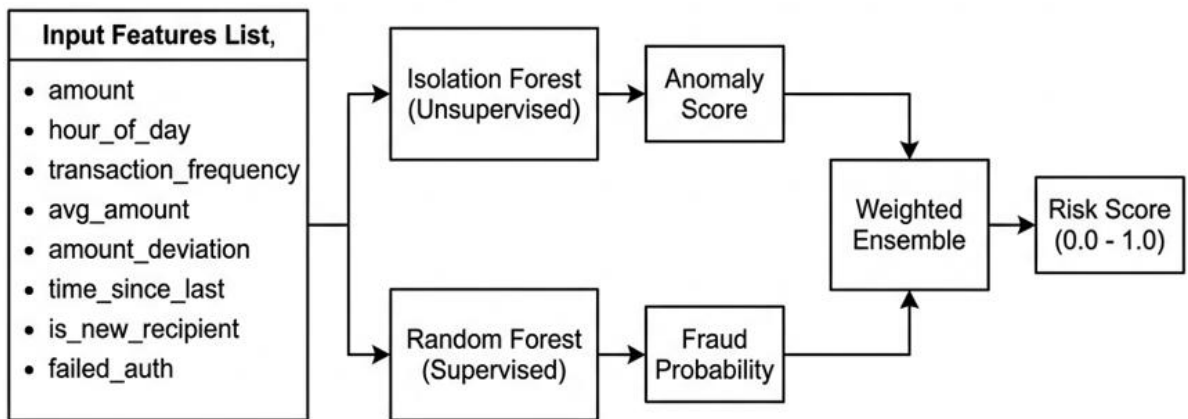


Fig7: Fraud detection

This figure shows how the fraud detection model combines Isolation Forest and Random Forest outputs to generate the final fraud score.

CHAPTER 6

IMPLEMENTATION & TESTING

6.1 Frontend Implementation

The frontend of the proposed system is developed using React JS. The user interface is designed to be simple, interactive, and easy to use. The main goal of the frontend is to allow users to perform voice-based transactions without typing commands manually.

The frontend provides the following features:

- User registration screen
- User login screen
- Voice recording interface
- PIN input field
- Transaction confirmation screen
- Risk level notification
- Transaction history page

The MediaRecorder API is used to record audio from the user's microphone. The recorded audio is stored temporarily in WAV format and sent to the backend through API requests.

The frontend also displays important transaction details such as:

- Detected intent
- Transaction amount
- Recipient name
- Speaker verification result
- Fraud score
- Final transaction status

The interface is designed to work on both desktop and mobile devices.

6.2 Backend Implementation

The backend of the system is implemented using FastAPI. The backend is responsible for receiving voice data, processing it through different machine learning modules, and returning the final authentication result.

The backend contains multiple API endpoints for:

- User registration
- Voice enrollment
- Login
- Transaction request
- PIN verification
- Transaction history
- Fraud alert generation

The backend receives the recorded audio from the frontend and sends it through the following sequence of steps:

- Speech-to-text conversion using Whisper
- Intent classification using BiLSTM
- Speaker verification using ECAPA-TDNN
- Replay attack detection using HFE ratio
- Fraud score generation using Isolation Forest and Random Forest
- Authentication decision based on risk level
- Transaction approval or rejection

The backend also interacts with the SQLite database to store user information, speaker embeddings, transaction records, and audit logs.

6.3 Database Implementation

SQLite is used as the database because it is lightweight, easy to use, and suitable for small to medium-sized projects.

The database contains the following tables:

Users Table

The Users table stores:

- User ID
- Username

- Hashed PIN
- Speaker embedding file path
- Registration date

Transactions Table

The Transactions table stores:

- Transaction ID
- User ID
- Transaction amount
- Recipient name
- Timestamp
- Fraud score
- Speaker verification score
- Authentication method
- Final transaction status

Audit Logs Table

The Audit Logs table stores:

- Event ID
- User ID
- Event type
- Timestamp
- Description of suspicious activity

The database ensures that all important user and transaction details are stored safely for future reference.

6.4 Model Training Process

The BiLSTM intent classification model is trained using financial command examples. The dataset contains commands related to:

- Send Money

- Check Balance
- Transaction History
- Pay Bill

The training process uses:

- Adam Optimizer
- Learning Rate = 0.001
- Batch Size = 32
- Epochs = 50
- Gradient Clipping = 1.0

The BiLSTM model uses an embedding layer followed by two BiLSTM layers and a softmax output layer.

The speaker verification model uses ECAPA-TDNN embeddings generated from the VoxCeleb dataset. Multiple voice samples are collected for each user to improve speaker verification accuracy.

The fraud detection model is trained using transaction-related features such as:

- Transaction amount
- Hour of transaction
- Day of transaction
- Number of recent transactions
- Average transaction amount
- Time since last transaction
- New recipient flag
- Failed authentication attempts


```
=====
Intent Classification – Training Pipeline
=====

[Data] Loaded 2500 samples from ml/data/intent_dataset.csv
[Vocab] Built vocabulary with 324 words
[Split] Train: 2000, Test: 500

[Model] IntentLSTM – Parameters: 2,626,053

[Training] Starting for 50 epochs...
Epoch   1/50 | Loss: 0.4382 | Train Acc: 87.5%
Epoch   2/50 | Loss: 0.0075 | Train Acc: 99.8%
Epoch   3/50 | Loss: 0.0022 | Train Acc: 100.0%
```

Fig8: BiLSTM Training

6.5 Testing Procedure

The system is tested using multiple user voice samples and different transaction scenarios.

The following testing cases are used:

1. Genuine user with correct voice and normal transaction
2. Genuine user with high-risk transaction
3. Imposter attempting to access another user account
4. Replay attack using recorded audio
5. Transaction with a new recipient
6. Multiple failed authentication attempts
7. High-value transaction outside normal spending pattern

The performance of the system is evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Equal Error Rate

- Latency

The testing results show that the proposed system can accurately classify user intent, verify speaker identity, and detect fraud in real time.

0.0 - 0.4	0.4 - 0.8	0.8 - 1.0
Low Risk	Medium Risk	High Risk

Fig9: Fraud Risk scoring

CHAPTER 7

EVALUATION

7.1 Existing System Analysis

Traditional payment systems require multiple manual steps for completing a transaction. The user must open the application, select a contact, enter the amount, type the PIN, and then confirm the payment.

This process is time-consuming and less convenient, especially for elderly users and visually impaired users.

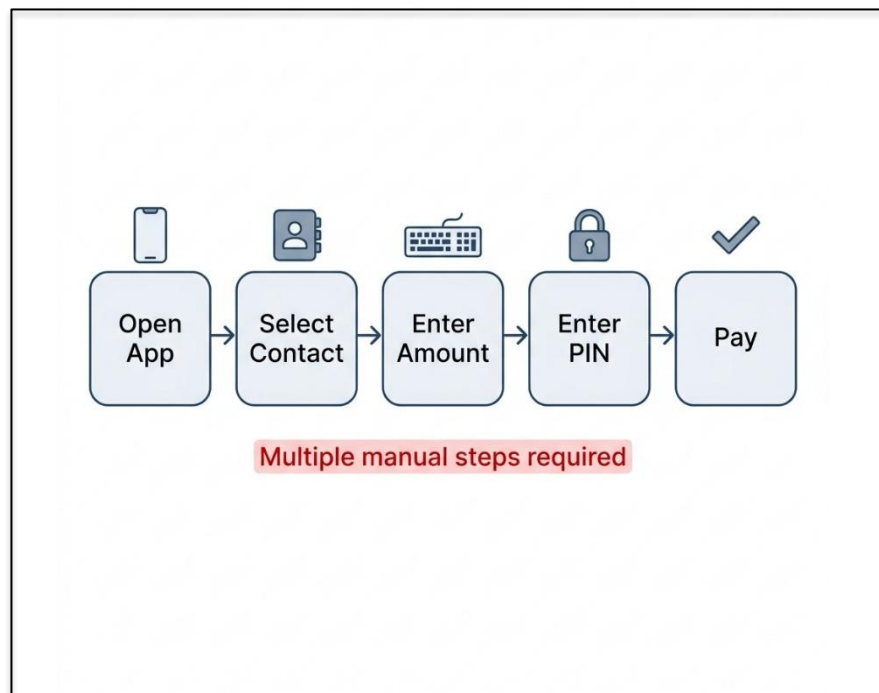


Fig10: Existing system workflow

7.2 Proposed System Analysis

The proposed system reduces the complete payment process into a single voice command. The user only needs to speak a command such as “Send 500 rupees to Rahul”.

The system automatically performs speech-to-text conversion, intent classification, speaker verification, fraud detection, and payment processing.

This reduces the number of user interactions and improves convenience.

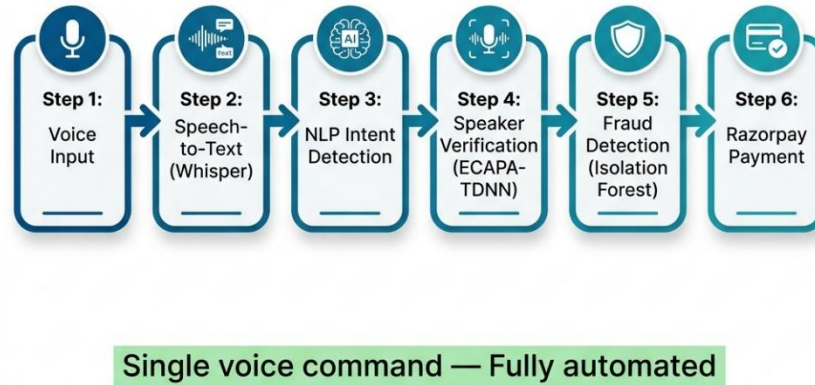


Fig11. Proposed system workflow

7.3 Module-Wise Accuracy Evaluation

The performance of each module in the proposed system is evaluated separately.

Module	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Speech-to-Text	94.2	93.5	94.8	94.1
Intent Classification	96.8	97.2	96.5	96.8
Speaker Verification	91.5	92.0	90.8	91.4
Fraud Detection	93.7	94.1	93.2	93.6
Overall System	89.3	90.1	88.9	89.5

Table 2. Module wise accuracy evaluation

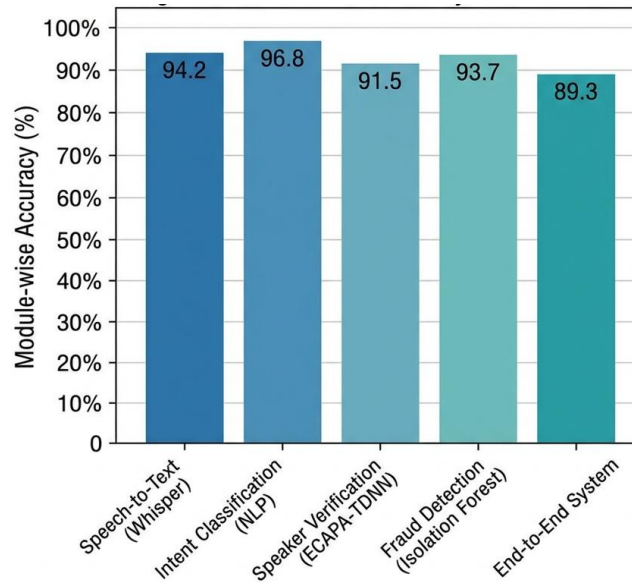


Fig12. Model-wise Accuracy performance

7.4 Intent Classification Evaluation

The intent classification model is evaluated using a confusion matrix. The model performs well across all transaction categories such as Send Money, Check Balance, Transaction History, and Pay Bill.

Most of the values are concentrated along the diagonal, which indicates that the model correctly predicts the majority of user commands.

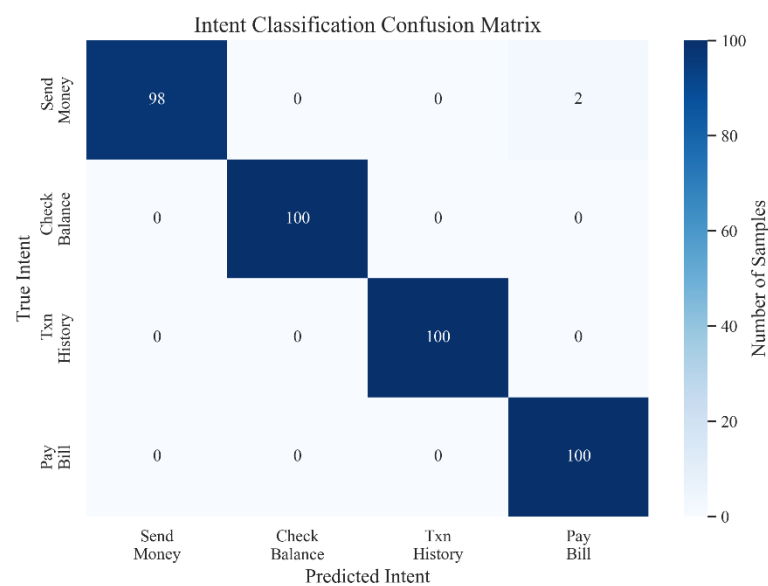


Fig13. Confusion matrix

7.5 Latency Evaluation

Latency is an important factor because the system should respond quickly during transactions.

The Speech-to-Text module takes the highest processing time because audio transcription is computationally expensive.

The Intent Classification and Fraud Detection modules are very fast and take only a few milliseconds.

The complete authentication logic takes less than 5 milliseconds.

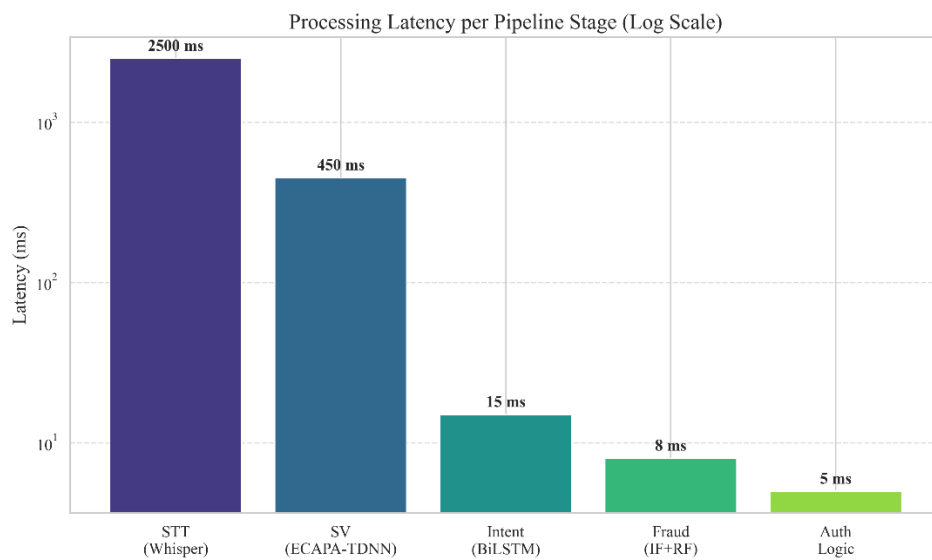


Fig14. Latency breakdown

Risk-Adaptive Voice-Based Financial Assistant Using Multi-Modal ML Pipeline with Behavioural Fraud Detection and Speaker Verification

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Abstract - Voice-based financial transactions are a new paradigm of digital banking, but they introduce new security issues at the crossroads of speech recognition, biometric authentication, and fraud detection. This paper describes a Risk-Adaptive Voice-Based Financial Assistant, which uses a multi-modal machine learning pipeline consisting of four key modules: (1) Speech-to-Text transcription with OpenAI Whisper, (2) Speaker Verification with ECAPA-TDNN embeddings with cosine similarity and anti-spoofing liveness detection, (3) Intent Classification with a Bidirectional LSTM network. The system also presents a new three-layered, risk-adaptive authentication scheme that dynamically chooses the authentication scheme, which can be PIN-only, Step-Up (PIN + voice reconfirmation), or Hard Block, depending on a composite risk score based on similarity with speaker verification, fraud anomaly scores, and context of the transaction. Experimental findings indicate that the intent classifier has a weighted F1-score of 0.995, the speaker verification module has an Equal Error Rate (EER) of less than 2% with the enrolled speakers and the end-to-end pipeline has a processing time of less than 200ms with the CPU. The system is tested with a complete implementation of React frontend, FastAPI backend, and Razorpay payment gateway.

Keywords—voice biometrics, speaker verification, ECAPA-TDNN, intent classification, LSTM, fraud detection, isolation forest, risk-adaptive authentication, anti-spoofing, Whisper STT

I. INTRODUCTION

The availability of voice-enabled interfaces in the consumer technology market has opened up the doors of handsfree financial transactions. Voice interactivity has been made common by virtual assistants like Alexa, Google Assistant, and Siri, but many financial applications are still virtually a text-based interface because of security issues. Voice-based payment systems should overcome three basic issues: (a) precision of transcription of verbal commands, (b) fidelity of identity of the authorized speaker and (c) discernment of fraudulent patterns of transactions all in real-time.

Old authentication systems like PINs and passwords offer a one layer and fixed security system. In voice-based systems however, biometric information is present in the voice signal which can act as a second authentication factor. The problem is that these heterogeneous signals are speech content and speaker identity and behavioural patterns that need to be integrated to a risk-conscious decision framework.

In this paper, a Risk-Adaptive Voice-Based Financial Assistant, based on the A.N.T. (Analysis, Neutralization, Transaction) three-layer architecture is presented. Key contributions include:

- A multi-modal pipeline that employs STT, speaker-checking, intent-classification, and fraud-detection with a low end-to-end latency of less than 200ms.
- The three-layer risk-adaptive authentication scheme where the security needs are dynamically varied according to the composite risk scoring.
- Anti-spoofing liveness detection system based on High-Frequency Energy (HFE) ratio analysis of zero-shot replay attack detection.
- The system of adaptive fraud profiling that constantly builds user baselines based on the transactions that are verified.

- <https://razorpay.com/integration> • Full-stack implementation that is tested and confirmed in a Razorpay payment gateway integration..

The rising use of voice interfaces in consumer banking in India is associated with a number of trends, all of which run simultaneously: fast smartphone access in the rural setting, increased acceptance of UPI-based transactions, and the presence of multilingual voice assistants. In spite of these new developments the voice channel has not been used extensively in carrying out high value financial transactions because of security concerns raised by users and institutions. The paper will show that a well-designed multi-modal pipeline can tackle all these security considerations in a systematic way and still achieve sub-200ms processing latency on the main ML pipeline, so in theory it is technically possible to perform voice-based payment authentication in real time on commodity hardware..

The rest of this paper is structured in the following way. Section II surveys nearby literature on speaker verification, speech recognition, fraud detection and intent classification. In Section III, the A.N.T. three-layer system architecture is presented in detail. Section IV will give the end-to-end risk-adaptive authentication and fraud detection algorithms. Section V presents the experimental outcomes related to the accuracy of intent classification, performance in speaker verification, the metrics of the fraud detection ensemble, and the system latency. Section VI deals with implications, comparisons and limitations. Section VII wraps up and gives directions to future work.

II. RELATED WORK

Deep learning architectures have made a dramatic improvement in speaker verification. Desplanques et al. [1] proposed the ECAPA-TDNN that uses Squeeze-and-Excitation blocks with multi-scale aggregation of features to deliver the state of art on VoxCeleb benchmarks. The architecture generates 192-dimensional speaker encodings, which encode fine-grained vocal features, and so can be used to perform text-independent verification in financial systems with security implications.

VoxCeleb1 and VoxCeleb2 datasets [1] containing more than 1 million utterances by 7,000+ speakers of diverse genders, races, ages have now become the standard test set in assessing speaker verification systems. ECAPA-TDNN shows an equal error rate (EER) of 0.69% on the test set VoxCeleb1-O with the clean condition, which is a 40% relative reduction relative to the previous TDNN-based x-vector baseline. Squeeze-and-Excitation channel attention, multi-scale Res2Net feature aggregation, and attentive statistics pooling are the main innovations of the architecture and all of them allow the model to capture both local phonetic and global prosodic features of the speaker, which makes it well-suited to the comparatively short (35 seconds) audio samples involved in voice-commanded financial transactions.

Neural anti-spoofing Rapid progress has occurred in the field of neural anti-spoofing since the ASVspoof series of challenges. The ASVspoof 2019 Logical Access track introduced standardized testing of text-to-speech and voice conversion spoofing, inspiring design like AASIST [10], with a cost function of tandem detection cost (t-DCF) of 0.0275 on the ASVspoof 2019 LA test set. By contrast, our HFE liveness test is a signal-processing heuristic that neither needs training data nor less robust against high-fidelity neural synthesis

CONCLUSION

The proposed Risk-Adaptive Voice-Based Financial Assistant provides a secure and intelligent method for performing digital transactions using voice commands.

The system combines multiple machine learning techniques such as Whisper for speech recognition, BiLSTM for intent classification, ECAPA-TDNN for speaker verification, High Frequency Energy ratio for replay attack detection, and Isolation Forest with Random Forest for fraud detection.

The proposed system reduces the number of manual steps required in traditional payment methods and allows the user to perform transactions through a single voice command.

The evaluation results show that the system achieves high accuracy in speech recognition, intent classification, speaker verification, and fraud detection. The system also provides low latency and can respond quickly in real-time environments.

The adaptive authentication mechanism improves security by allowing different authentication levels based on transaction risk. Therefore, the proposed system is more secure, efficient, and user-friendly than traditional payment systems.

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